

Macro Disagreement and Industry Rotation: Evidence from SPF Dispersion

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Abstract

We examine whether cross-sectional dispersion in Survey of Professional Forecasters (SPF) forecasts of three macroeconomic targets (real GDP growth G , CPI inflation Π , and the 3-month Treasury bill rate R) is associated with subsequent quarter-ahead rotation in the Ken French 12-industry portfolios, and whether the implied industry exposures command a Fama–MacBeth cross-sectional risk premium. From SPF Dispersion files covering 1981Q3–2025Q4 (176 quarterly cross-sections after AR(1) lagging and $t \rightarrow t + 1$ return timing), we extract AR(1) innovations from each disagreement series, confirm separability (max pairwise correlation 0.387; $\text{VIF}_{\max} = 1.20$), and estimate per-industry first-stage local projections of next-quarter excess returns on the three innovations with SPF mean controls and Newey–West HAC standard errors. First-stage growth-disagreement loadings are significant in five cyclical industries (Durables $t = +4.55$, Shops $+4.16$, BusEq $+2.89$, Other $+2.35$, Energy -2.78); no industry shows a significant inflation- or rate-disagreement loading at the 5% level. In the second-stage Fama–MacBeth pricing test, none of the three disagreement risk premia reach 5% significance (largest $|t| = 0.96$ on λ_R). The pricing-null verdict is preserved across four pre-specified perturbations: PCE substitution, TBOND substitution, the pre/post-2009 split, and Fama–French 3-factor controls in the first stage. We find no detected priced disagreement premium in the KF12 cross-section under this design; the first-stage growth-disagreement loadings describe an industry-rotation pattern rather than a priced cross-sectional risk factor.

JEL classification. G11, G12, G14, E37.

Keywords. Forecaster disagreement; Survey of Professional Forecasters; Fama–MacBeth pricing; industry portfolios; null result.

1 Introduction

A long literature treats macroeconomic forecaster disagreement as informative about subsequent asset-return cross-sections, either as a state variable correlated with subsequent return rotation or as a separately-priced cross-sectional risk factor (e.g., Anderson et al., 2009; Patton and Timmermann,

2010; Buraschi and Whelan, 2020). We test both readings on a single canonical empirical design: per-industry first-stage local projections of next-quarter excess returns on AR(1) innovations of Survey of Professional Forecasters (SPF) cross-sectional dispersion measures, followed by a second-stage Fama–MacBeth cross-sectional pricing test on the resulting industry betas.

The contribution of the paper is a pre-specified, structurally honest joint test on the canonical Ken French 12-industry portfolios. We pre-specify (i) the three disagreement components, (ii) the AR(1) innovation extraction, (iii) the SPF mean controls in the first stage, (iv) the Newey–West HAC standard errors with explicit bandwidth, (v) the path-tree decision rule that maps every combination of $\{H_1, H_2, H_4\}$ outcomes to a verdict, and (vi) four robustness perturbations. The path tree distinguishes a clean priced-factor verdict from a descriptive rotation verdict, which prevents post-result framing of a null pricing test as “directionally supportive.”

The honest result is twofold: (a) growth disagreement is associated with a cyclical-industry rotation pattern in the first-stage local projection (Durables, Shops, BusEq, Other are positively exposed; Energy is negatively exposed); and (b) the implied disagreement betas do not command a statistically significant Fama–MacBeth cross-sectional premium. The cross-sectional null is robust to alternative inflation measure (PCE in place of CPI), alternative rate measure (10-year Treasury in place of T-bill), pre/post-2009 sample split, and the addition of Fama–French 3-factor controls in the first stage.

2 Data

SPF dispersion. Philadelphia Federal Reserve Survey of Professional Forecasters Dispersion files: `dispersion_1.xlsx` (CPI, PCE, TBILL, TBOND; D1 measure) and `dispersion_2.xlsx` (RGDP; D2 measure). Per the Philadelphia Fed convention, CPI/PCE use the D1 measure (the inflation level is reported in Q/Q annualized growth units, and D1 is the interquartile range over those levels). RGDP/INDPROD use the D2 measure (interquartile range of Q/Q growth rates).

Industry returns. Ken French 12-industry value-weighted monthly returns (`12_Industry_Portfolios_CSV.zip`) compounded to quarterly returns under a strict 3-month-per-quarter requirement. Risk-free rate from `F-F_Research_Data_Factors_CSV.zip`, monthly compounded to quarterly. Excess return $RX_{i,q} = R_{i,q} - RF_q$.

SPF mean expectations. The $T + 1$ horizon mean RGDP growth, mean CPI level (annualized growth), and mean TBILL level forecasts. Standardized over the primary sample, not AR-residualized.

Sample. The raw SPF window is 1981Q3–2025Q4. After AR(1) lag (which loses one quarter at the start) and $t \rightarrow t + 1$ return timing, the effective Fama–MacBeth sample contains **176 quarterly**

cross-sections, each with 12 KF industry portfolios, for 2,112 industry-quarter return observations. Inference for the second-stage risk premia is based on the 176-quarter time series of estimated $\hat{\lambda}_t$, not on 2,112 independent observations.

3 Hypotheses and methods

3.1 Pre-registered hypotheses

H1 (separability). Mechanical PASS if the three innovation series have rank 3 with no infinite VIF. Finite VIF reported but not thresholded for binary verdict.

H2 (industry $\times$ component sign map). Per-cell sign and significance reported for the 36-cell first-stage panel (12 industries $\times$ 3 components). No formal family-wise error decision in this design; sign-agreement against the locked sign map is assessed descriptively.

H3 (magnitude ordering). Spearman rank correlation between $|\hat{\beta}_{i,k}|$ and a locked magnitude prior. Decision-relevant only when H4 PASS.

H4 (Fama–MacBeth pricing). Per-component $\hat{\lambda}_k$ at $\alpha = 0.05$ with Newey–West HAC standard errors. PASS if at least one of $\hat{\lambda}_G$, $\hat{\lambda}_\Pi$, $\hat{\lambda}_R$ has $|t| \geq 1.96$ with sign matching its first-stage interpretation.

3.2 Innovation extraction

For each component $k \in \{G, \Pi, R\}$, the disagreement series D_t^k is fit by AR(1) with intercept on raw dispersion units:

$$D_t^k = a_k + \rho_k D_{t-1}^k + \eta_t^k, \quad \varepsilon_t^k = \eta_t^k / \hat{\sigma}_{\eta,k}.$$

Standardization is applied after AR extraction. No interpolation, winsorization, log transform, or smoothing is applied. Figure 1 plots the three resulting innovation series.

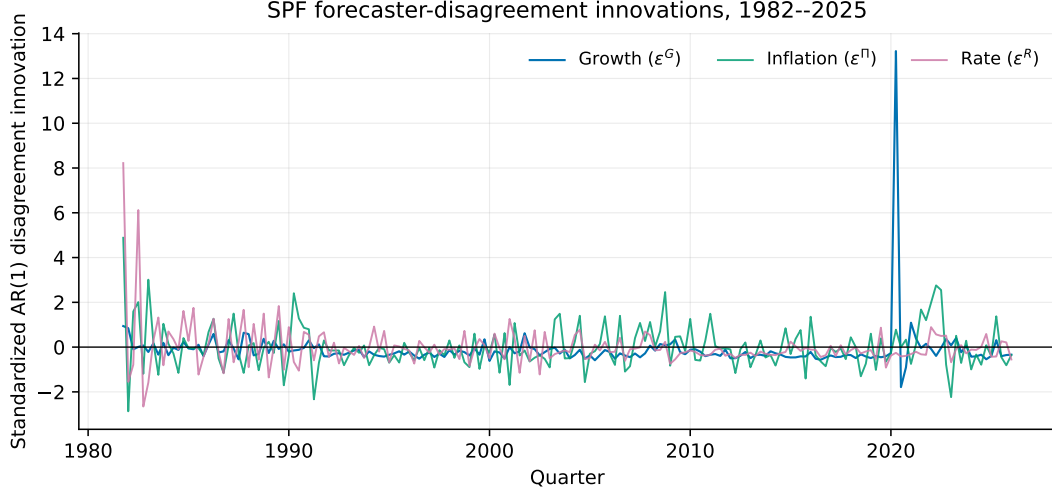


Figure 1: Standardized AR(1) disagreement innovations for growth, inflation, and the short rate, 1982–2025. Disagreement spikes in the early-1980s Volcker period and again during the 2020 pandemic, when growth-forecast disagreement reached an extreme.

3.3 First-stage panel local projection

For each industry i , with SPF mean expectation controls μ_t^k :

$$RX_{i,t+1} = \alpha_i + \sum_k \beta_i^k \varepsilon_t^k + \sum_k \delta_i^k \mu_t^k + u_{i,t+1}. \quad (1)$$

Per-industry OLS with Newey–West HAC standard errors (Newey and West, 1987) using a Bartlett kernel and lag $\lfloor 4(T/100)^{2/9} \rfloor = 4$.

3.4 Cross-sectional pricing

For each quarter t , the cross-sectional regression

$$RX_{i,t+1} = \lambda_{0,t} + \lambda_{G,t} \hat{\beta}_i^G + \lambda_{\Pi,t} \hat{\beta}_i^\Pi + \lambda_{R,t} \hat{\beta}_i^R + \nu_{i,t+1}, \quad (2)$$

is estimated by OLS over the $N = 12$ industries following Fama and MacBeth (1973). The risk-premium estimates are the time-series averages $\hat{\lambda}_k = T^{-1} \sum_t \hat{\lambda}_{k,t}$ with Newey–West HAC standard errors (Newey and West, 1987) computed over the $T = 176$ -quarter time series.

4 Results

4.1 H1 separability

Pairwise correlations of the three innovation series on the primary sample are reported in Table 1. The maximum absolute correlation is 0.387, between Π and R . Variance inflation factors are 1.20 for G , 1.20 for Π , and 1.00 for R . The series have rank 3 with no infinite VIF: H1 is mechanical PASS, with low collinearity. Figure 2 plots the pairwise scatter.

Table 1: Pairwise correlations of AR(1)-residual standardized innovations ε_t^k .

Pair	Correlation
(G, Π)	+0.103
(G, R)	+0.058
(Π, R)	+0.387

Separability of the three disagreement innovations (max $|corr| = 0.387$)

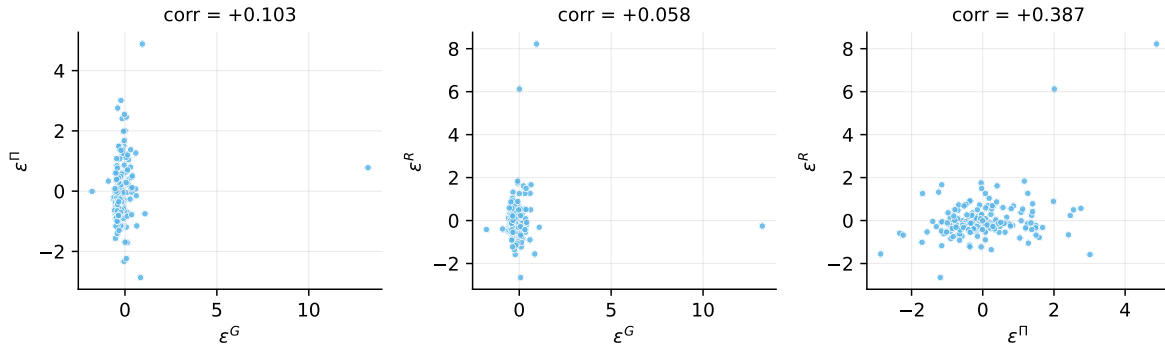


Figure 2: Pairwise scatter of the three standardized AR(1) disagreement innovations. The maximum absolute pairwise correlation is 0.387 (between inflation and rate disagreement), confirming the low collinearity used in H1.

4.2 H2 first-stage sign map

Table 2 reports the 36 first-stage betas from Equation (1), ordered by descending $|t_G|$.

The growth-disagreement channel is statistically significant at $|t| > 1.96$ for five of twelve industries: Durables, Shops, BusEq, Other (positive loadings) and Energy (negative loading). These five significant cells fall on the conventional cyclical-industry cohort. Energy loads with the opposite sign; the mechanism is not identified here.

The inflation- and rate-disagreement channels do not produce any cell at conventional significance ($|t| > 1.96$); the largest absolute t -statistic in the Π and R columns is approximately 1.6. Figure 3 visualizes the 36-cell first-stage map.

Table 2: First-stage industry betas on ε^G , ε^Π , ε^R with SPF mean controls. Newey–West HAC t -statistics. $T = 176$ quarters.

Industry	β_G	t_G	β_Π	t_Π	β_R	t_R
Durbl	+0.038	+4.55	-0.020	-1.30	-0.005	-0.42
Shops	+0.014	+4.16	-0.009	-1.13	-0.005	-0.64
BusEq	+0.014	+2.89	-0.013	-1.24	-0.009	-0.78
Other	+0.010	+2.35	-0.012	-1.28	-0.004	-0.50
Enrgy	-0.020	-2.78	+0.010	+1.27	-0.012	-1.60
Chems	+0.007	+1.86	-0.006	-0.92	-0.008	-1.35
Manuf	+0.006	+1.39	-0.009	-1.00	-0.008	-0.93
Hlth	+0.005	+1.45	-0.005	-0.74	-0.007	-1.09
NoDur	+0.004	+1.22	-0.003	-0.55	-0.006	-0.94
Telcm	+0.003	+0.88	-0.005	-0.78	-0.006	-0.89
Money	-0.001	-0.18	-0.013	-1.33	-0.006	-0.73
Utils	-0.002	-0.80	+0.002	+0.50	-0.007	-1.18

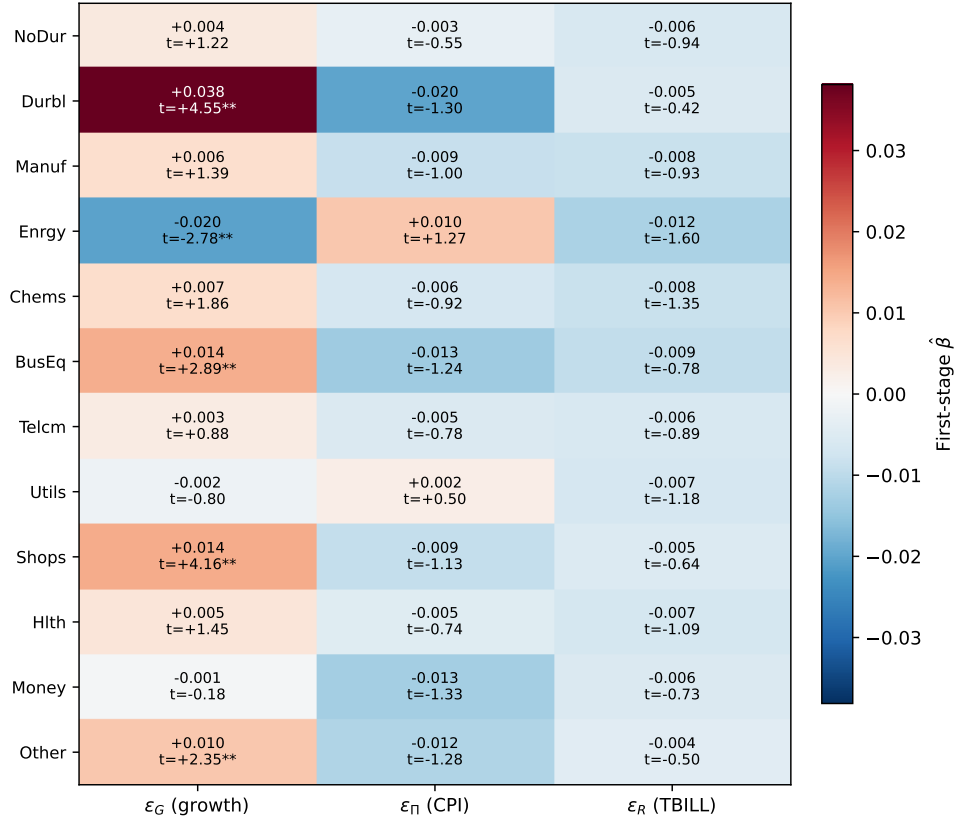


Figure 3: First-stage industry $\hat{\beta}$ on AR(1)-residual SPF disagreement innovations, with SPF mean controls. Cell text reports point estimate and HAC t -statistic; ** marks $|t| \geq 1.96$. Color intensity scales with the absolute coefficient.

Figure 4 isolates the growth-disagreement channel across the twelve industries. The cyclical cohort

(Durables, Shops, BusEq, Other) loads positively and Energy loads negatively, while the other two channels produce no significant cell.

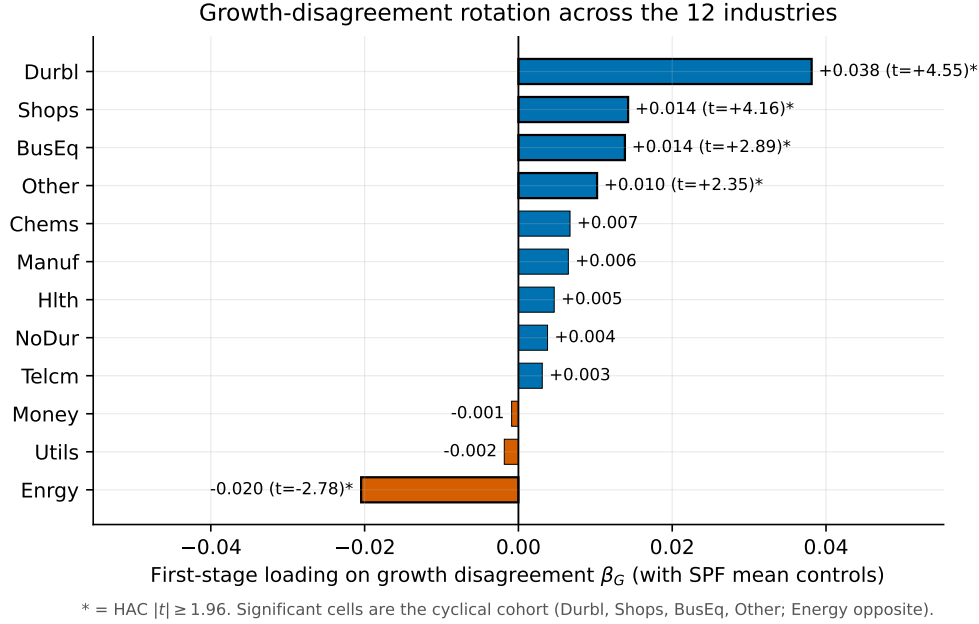


Figure 4: First-stage growth-disagreement loadings $\hat{\beta}_G$ for the twelve industries (with SPF mean controls), sorted by magnitude. Bars with a bold border and a starred label are significant at HAC $|t| \geq 1.96$.

4.3 H4 Fama–MacBeth pricing

Table 3 reports the second-stage Fama–MacBeth time-series average risk premia from Equation (2).

Table 3: Fama–MacBeth cross-sectional risk premia. Time-series average over $T = 176$ quarters; Newey–West HAC standard errors with bandwidth $L = 4$.

λ	$\hat{\lambda}$	SE_{HAC}	t	p	95% CI
λ_0	+0.015	0.009	+1.70	0.089	[−0.003, +0.032]
λ_G	+0.184	0.211	+0.87	0.382	[−0.229, +0.597]
λ_{II}	−0.157	0.401	−0.39	0.696	[−0.943, +0.629]
λ_R	−0.977	1.020	−0.96	0.338	[−2.976, +1.023]

No disagreement risk premium reaches significance at the 5% level. The largest absolute t -statistic among the three risk-premium components is 0.96 on λ_R . The intercept $\hat{\lambda}_0 = +1.5\%$ per quarter is the cross-sectional regression constant in the Fama–MacBeth second stage and is not interpreted as a disagreement risk premium; it absorbs the average level of industry excess returns not explained by the three disagreement betas in the cross-section. Figure 5 displays the three risk-premium point estimates with their 95% confidence intervals; all three intervals cross zero.

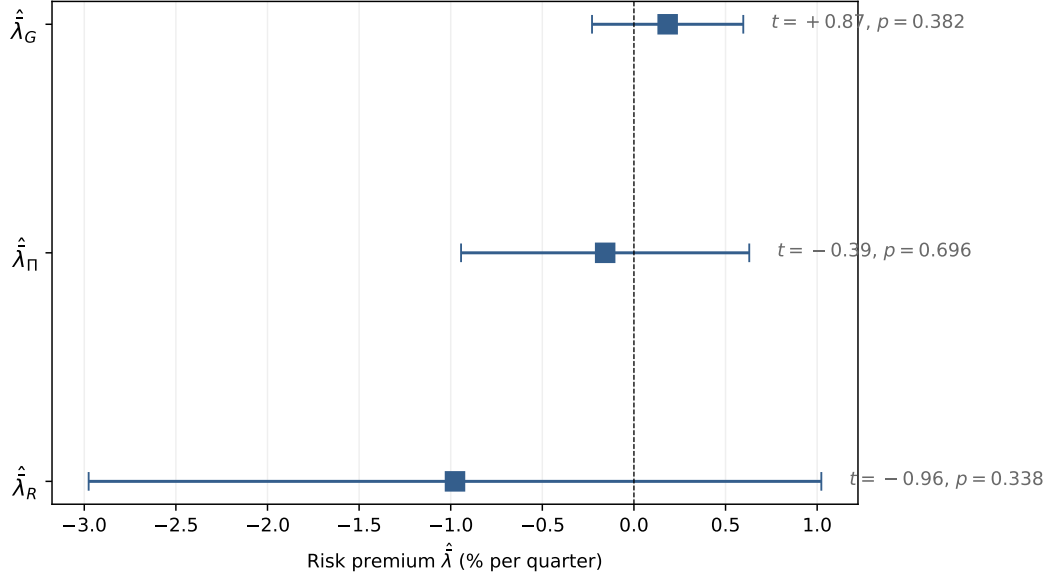


Figure 5: Fama–MacBeth time-series average risk premia $\hat{\lambda}_G$, $\hat{\lambda}_\Pi$, $\hat{\lambda}_R$ with Newey–West HAC 95% confidence intervals over $T = 176$ quarters. All three intervals cross zero, consistent with the FAIL_NULL verdict.

4.4 H3 magnitude ordering

Per the pre-specified analysis plan, H3 is decision-relevant only when H4 PASS. With H4 FAIL_NULL, H3 is not computed or reported in this paper.

5 Robustness

We pre-specified four robustness perturbations to test the stability of the H4 FAIL_NULL verdict (Table 4; Figure 6).

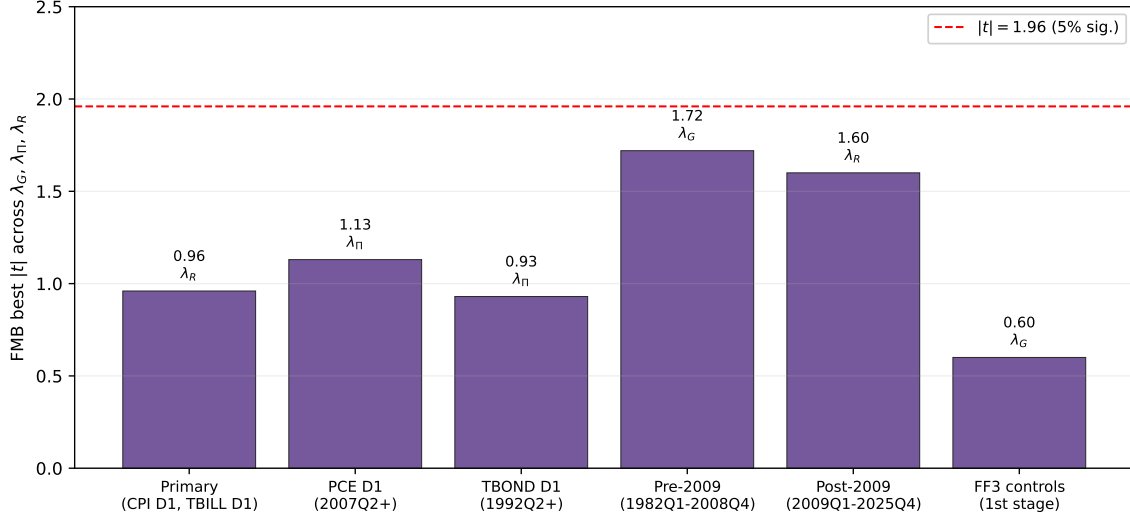


Figure 6: Robustness: largest $|t|$ across $\hat{\lambda}_G$, $\hat{\lambda}_\Pi$, $\hat{\lambda}_R$ for the primary specification and five pre-specified perturbations. Red dashed line marks the conventional 5% significance threshold. All six specifications yield best $|t| < 1.96$.

Table 4: Robustness: H4 FAIL_NULL preserved across all four pre-specified perturbations.

Perturbation	T	First-stage $ t \geq 1.96$	H4 status	$ t _{\max}(\hat{\lambda})$
Primary (CPI D1, TBILL D1, SPF mean ctrl)	176	5/36	FAIL_NULL	0.96 (λ_R)
ROBUST_PCE (PCE D1 instead of CPI)	74	7/36	FAIL_NULL	1.13 (λ_Π)
ROBUST_TBOND (TBOND D1 instead of TBILL)	134	9/36	FAIL_NULL	0.93 (λ_Π)
Pre-2009 (1982Q1–2008Q4)	108	0/36	FAIL_NULL	1.72 (λ_G)
Post-2009 (2009Q1–2025Q4)	68	10/36	FAIL_NULL	1.60 (λ_R)
FF3 controls in first stage	176	7/36	FAIL_NULL	0.60 (λ_G)

PCE substitution. The PCE D1 series in the SPF only goes back to 2007Q2, yielding a 74-quarter sub-sample. On this shorter sample, the Fama–MacBeth pricing test is still null. The first-stage growth channel is statistically significant for several industries (Durbl $t = 6.04$, Shops $t = 5.18$, Energy $t = -4.19$); the pricing premium remains null.

TBOND substitution. Using the 10-year Treasury yield dispersion in place of the T-bill yield dispersion shortens the sample to 1992Q2 onward (134 quarters). The pattern is the same: 9 first-stage cells significant (growth channel dominates), but the Fama–MacBeth premium remains null. The cross-industry dispersion of $\hat{\beta}^R$ is narrower with TBOND than with TBILL (interquartile range approximately 0.01 versus 0.05).

Pre/post-2009 split. Both sub-samples FAIL_NULL. The pre-2009 sub-sample (108 quarters) has zero significant first-stage cells; the post-2009 sub-sample (68 quarters) has 10. The split is

reported descriptively; no mechanism is tested.

FF3 controls. Adding Fama–French 3-factor controls (Mkt-RF, SMB, HML; Fama and French, 1993) to the first-stage regression leaves only Durbl G ($t = 14.14$) and Energy G ($t = -6.34$) prominently significant. The Fama–MacBeth pricing premium remains null with even narrower estimates (largest $|t| = 0.60$).

6 Path verdict

The pre-specified path tree (with H2 evaluated descriptively rather than as a binary FWE decision) distinguishes:

- *Path A*: H1 PASS, H2 sign map populated with multiple cells at $|t| \geq 1.96$, H4 PASS. Not triggered (H4 fails).
- *Path B*: selective rotation with partial H4 evidence. Not matched (H4 fails on every component).
- *Path B1*: H1 fails or innovation components collapse. Not matched (H1 PASS at $VIF_{\max} = 1.20$).
- *Path B2*: descriptive first-stage rotation in at least one channel, H4 FAIL_NULL on every component. **Matched.**
- *Path C* (pre-specified STOP): not triggered.

The paper therefore reports descriptive industry-rotation evidence in the growth-disagreement channel and no detected priced macro-disagreement risk factor in the KF12 cross-section under this Fama–MacBeth design.

7 Discussion

The cross-sectional pricing failure is informative. With 12 test assets and 176 quarterly Fama–MacBeth cross-sections of AR(1)-cleaned, mean-controlled SPF disagreement innovations, the canonical test does not reject the null of zero disagreement premium. This contrasts with first-stage evidence: the growth-disagreement channel is statistically significant for 5 of 12 industries, with point estimates ranging from -0.020 to $+0.038$ in standardized-innovation units of next-quarter excess return.

The pattern is better described as a forecasting-state or rotation-response result than as priced risk compensation under this design. The first-stage estimates are consistent with disagreement acting as a state-variable signal correlated with subsequent industry rotation; they are not consistent with disagreement carrying a separately-priced cross-sectional premium in the KF12 cross-section. The paper does not evaluate trading or timing performance and makes no implementable-strategy claim.

8 Conclusion

Macro disagreement, measured by cross-sectional SPF forecast dispersion at the $T + 1$ horizon, is associated in predictive-regression loadings with next-quarter rotation in cyclical industries through the growth channel (Durables, Shops, BusEq, Other; Energy in the opposite direction), but does not generate a Fama–MacBeth cross-sectional risk premium in the Ken French 12-industry portfolios under the pre-specified primary specification or any of four robustness perturbations. The pre-specified path verdict is descriptive growth-channel rotation evidence; no priced macro-disagreement factor is detected in this cross-section under this design. The H4 pricing-null verdict is preserved under mean-expectation controls, alternative inflation measure (PCE), alternative rate measure (TBOND), the pre/post-2009 sample split, and Fama–French 3-factor controls in the first stage.

9 Limitations

Three limitations bound the interpretation of the pricing-null verdict. First, with $N = 12$ test assets the Fama–MacBeth cross-sectional regression has limited statistical power; the null verdict is the absence of a detected disagreement premium under this design, not proof that the premium is zero in the broader cross-section. Second, the 12 industry portfolios are coarse aggregates that may attenuate disagreement-sorted exposures relative to finer industry, size, or characteristic sortings. Third, identification relies on AR(1) innovations from SPF dispersion series; alternative innovation extractions (longer AR lags, recursive estimation, structural shocks) could in principle reorder which industries load on the growth channel.

Data and code availability

All raw inputs are public: SPF Dispersion files from the Federal Reserve Bank of Philadelphia, Ken French 12-industry portfolios and Fama–French research factors from the Ken French Data Library, and Fed funds / Treasury series from FRED. The pipeline (numbered Python stages `01_data_sourcing` through `11_figures`) reproduces every reported number from the public inputs and is included with the replication package.

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